

The Guarded Agent

An autonomous options paper-trading agent where AI proposes - and deterministic controls decide.

ALPACA AI TRADING AGENTS HACKATHON

AI

evidence + thesis



RISK

gates + bounded loss

PROPOSAL → DECISION

An AI can explain a trade - and still lose the guardrail.

THE FAILURE MODE

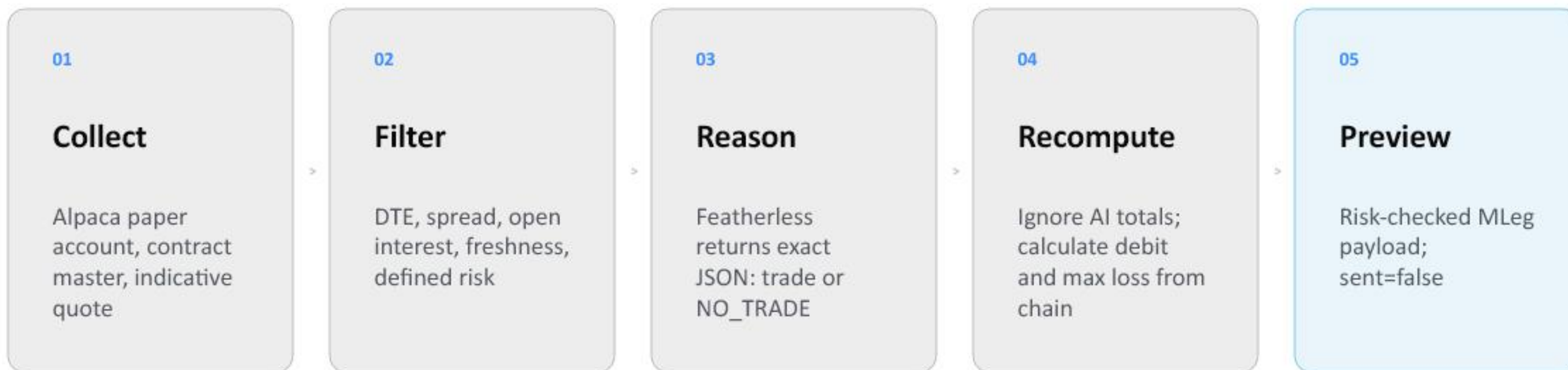
- Model output is probabilistic.
- Option quotes can be stale or wide.
- A convincing thesis is not a risk limit.

THE DESIGN RESPONSE

- AI sees sanitized, timestamped evidence.
- The engine recomputes debit and maximum loss.
- Only defined-risk options can reach a preview.

The model is a research layer - never a broker client.

The same evidence travels through five explicit safety boundaries.



No provider failure, malformed response, stale data, or audit failure can become an order.

Every candidate is options-only, defined-risk, and independently measurable.

The AI may choose the structure; market evidence supplies the economics.

STRATEGY	USE	RECOMPUTED	GATE EXAMPLES
Long call	Convex upside	$\text{Ask} \times 100 \times \text{qty}$	$\text{DTE} \cdot \text{spread} \cdot \text{OI}$
Long put	Convex downside	$\text{Ask} \times 100 \times \text{qty}$	$\text{DTE} \cdot \text{spread} \cdot \text{OI}$
Call debit spread	Bullish, lower debit	$(\text{long ask} - \text{short bid}) \times 100 \times \text{qty}$	$\text{Width} \cdot \text{freshness} \cdot \text{portfolio risk}$
Put debit spread	Bearish, lower debit	$(\text{long ask} - \text{short bid}) \times 100 \times \text{qty}$	$\text{Width} \cdot \text{freshness} \cdot \text{portfolio risk}$

No naked short options. No undefined loss. Abstention is a valid decision.

The account is clean - and the broker is still behind a gate.

\$100,000 verified starting equity	LEVEL 3 options approval / trading	0 orders · fills · positions	OFF execution flag
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What the deterministic gates check before a preview

- Per-trade and total portfolio risk
- Daily drawdown and open-position count
- Bid/ask width, open interest, and DTE

PAPER ONLY

The public demo shows a preview.
It does not send a broker order.

What the demo proves - and what it does not.

VALIDATED NOW

- Paper account health and fresh-account state
- Alpaca MCP option chain, quote, IV, and Greeks reads
- Strict AI schema, budget guard, and hash-chained audit
- Four-strategy scenario simulator with fixed seeds

CURRENT GO / NO-GO

NO-GO

Execution remains disabled until historical validation, live shadow evidence, and final dry-run review are complete.

Scenario simulation is not presented as a backtest.